

GROCERY STORE MANAGEMENT SYSTEM (SRS)

SOFTWARE REQUIREMENT SPECIFICATION

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SESSION: 2023-2025

COLLEGE: NATIONAL COLLEGE HASILPUR



DEPARTMENT OF COMPUTER SCIENCE



[Grocery Store management system]

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SESSION: 2023 - 2025 | <DEPARTMENT OF COMPUTER SCIENCE /NATIONAL COLLEGE HASILPUR>

Revision History

Date	Description	Author	Comments
<date>	<Version 1>	<NIMRA KOUSAR >	<First Revision>

Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

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CHAPTER 1: INTRODUCTION

Store is a place where access material is kept which will be used as and when required. Loss of items, deterioration, obsolescence and inadequacy are treated “as a part of life “store management system ”is to receive material” to protect them while in storage from damage& unauthorized removal to issue the material in the right quantities, at the right time to the right places and provide to these service promptly and at least cost.

1.1 Purpose:

The purpose of the document is to collect and analyze all the ideas. So we want to sort out this product will be used in to order in gain a better understanding of the project develops. In short, the purpose of this SRS document is providing a detailed overview of our software product its parameter and goals. This document describes the project’s audience and its user interface, hardware, software requirements. It defines its functionality. None the less, it helps to design software delivery life cycle processes.

1.2 Scope:

In this subsection:

- Minimize cost of production through minimize cost on materials.
- Maintain the value of materials.’
- Service to user departments.
- Establishing co-ordination with other departments.
- Advising materials manager.

1.3 Definitions, Acronyms, and Abbreviations:

- WWW (worldwide web set of large interlinked document on internet).
- SQL (Structure Query Language).
- GUI (Graphical User Interface).
- Modify (for modification in database or user record).

- Admin (Login for attraction with application).
- Logout (Logout for UN attraction with application).
- RESUME (To restarting something).
- PAUSE (To stop something).

1.4 References:

In this subsection

- (1) www.google.com.
- (2) www.draz.pk.
- (3) www.wikiadia.com
- (4) www.3school.com.

1.5 Overview:

This document provides a general description, including features of the projects. The products of hardware, data requirements and functional of the products. General description of the project is discussed in section 2 of this document. Section 3 will explain the detailed functional specification system and other requirements of the system.

CHAPTER 2: THE OVERALL DESCRIPTION

This document contains the problem statement that the current system is facing which are company's growth opportunities. It further includes a list of stakeholder and users of the proposed solution. It also illustrates the needs and requirements of stakeholders that were identified in the brainstorming exercise as part of the requirements of stakeholder workshop. It further list and brief describe each of the major features and a brief description of each proposed system.

The following SRS contains detail product perspectives from different stakeholders. It provides the details product functions of the store with the user characteristics of permissions and requirements subsets.

2.0.1 Initiation of order:

This step is involves receiving the order form customer, or receiving goods from mill which need to be sold.

2.0.2finding mills/customer:

System users then look for a mill from where they can find good that are required by customer. This step also includes agreements on different aspects for order like delivery, time, price etc

2.0.3 create/send Performa invoice:

The system will then create the Performa invoice to communicate with the customer. This is considered to be the main agreement between user and customer on price and other aspects of order. In some case the customer does not need to submit formal approval of Performa invoice.

2.0.4 Order to mill:

After the customer agrees on perform invoice, the system will order to the mill. This order will includes all detail of good, delivery time, shipment including ports details etc.

2.0.5 Handle shipment:

After customer agrees on Performa invoice, when good are manufactured by the mill. They transport those goods to the system user. This step also including finding shipping company, communication ports of delivery and goods details etc.

2.0.6 Generates documents for bank:

In many cases, to complete the delivery, the system user need to generate and compiles documents in order to let customer's t get delivery from port. This process includes communication between bank and user, the packing list of the finalizing and the bank/customer

for compiling documents.

2.0.7 Receive payment:

After completion of all the steps, the system will receive the payment on the bases of the system. In some cases, the user will generate the invoice in the end, after delivery has been done.

2.1 Product Perspective:

Put the product into perspective with other related products. If the product is independent and completely self-contained, it should be said here. If the SRS defines a product that is a major component of a component as it is often, then this subsection relates to the requirements of the longer system to the functionality of the software and interfaces that system and software. If you are building a real system compare its similarity and differences to other system in the market places. If you are research-oriented project, what related research compare to the system you are planning to build. A block diagram showing the major components of the larger system, interconnections, and external interfaces can be helpful. This is not a design or architecture picture. It is more to relevant to the context, especially if your system will interact with external actors. The system you are building should be shown as a black box. Let the design document present the internals. The following subsections describe how the software operates inside different constraints.

2.1.1 Operations:

- Admin must login for interaction
- Admin can also modify the record of database and user records
- Logout process is used for un-attraction from the desktop application
- It can maintain stock of goods
- It should be able to manage customer requirements.
- Customer can see detail of product.
- Customer can order by visiting the store.

2.1.2 Site Adaptation Requirement:

Store management system will attract with local SQL

2.2 Product Functions:

For clarity:

- Manager must login for attraction of disk application
- User visit this store and decide which one things want to buy
- Manager will save the record about user in database
- The database is automatically maintained.
- The software should be able to show the entire product and with price so that customer can select goods according to its needs and should be able to by it.
- *The* software should be able to show graphical entire phase to the customer so that he can be order to specification and quality.

2.3 User Characteristic:

The user is simple any one that has access to the internet and web browser. The assumed that user is familiar with the computer to operator the browser.

- User no needs any hardware.
- Having capacity to buy any product.
- Client cannot able to modify anything's.

- User cannot interact with this system.

2.4 General Constraints:

Provide a general description of any other items that will limit the developer's options. These can include:

- Graphical is interface design in English
- Manager need authentic requirement for interaction with the store management
- System must have capability of logout
- In authentication requirements, user name and password must be required.
- Time and cost.
- Scope.
- Quality.
- Benefits and Risk.
- Result.
- Resources.
- Activity performance.
- Consult your audiences.
- Review relevant written materials.

2.5 Assumptions and Dependencies:

This subsection of the SRS should list each of the factors that affect the requirements stated in the SRS. These factors are not design constraints on the software but are any changes to them that can affect the requirements in the SRS.

- First one assumption is that this is a desktop application.
- Does not need any extra hardware.
- System is integrated with SQL server.

CHAPTER 3: SPECIFIC REQUIREMENTS

- Maintained data associated with the inventory.
- Maintained record for many customers.
- Show a list of all available brands.
- Allow manager to update the stock quantity.

3.1 External Interface Requirements:

There is no need for any external software or hardware, the system is independent that can be run with no external support.

3.1.1 System Interfaces:

On the start of this application admin will authenticate his /her self. Then main view will open where all model of grocery are in drag drop box and other fields for buyer information gathering. After filling all fields and selecting items and clicking one button a confirmation message will show. You can see all database grocery stock on clicking check stock. Same like that on main Menu bar will exist for functionality of help checking stock sample images etc. Also main view consists on logout button for logout for the entire system.

3.1.2 Interfaces:

- As describe above main page will consist on same fields like name drag drop box etc. After filling it click on done when you will done stock or selecting item will less and buyer information will store in database etc.
- When manager will click on stock button then all stocks of grocery from database will show on one form.
- When you click on sample image icon from menu bar image of all grocery models with price will show.
- When user will click on logout all forms will close and just login will show on the screen.
- When manager give username and password and click on login the main view will open and login form will un-visible.

3.1.3 Hardware Interfaces:

- Intel core 2quad.
 - Memory 4 GB DDR3
 - Hard disk1*360 GB HDDSATA
 - Ethernet card 10/100/1000Mbps.
-
- Microsoft windows2003 small business server.

3.1.4 Software Interfaces:

- SQL as a database and OS (windows).
- Visual studio on OS (windows).

3.1.5 Communications Interfaces:

System will communicate with local SQL server.

3.2 Functional Requirements:

Our system provides these facilities.

- Store the inventory record.
- Store buyer record.
- Facility to modify in future.
- Store daily sale.
- Manager must login before use because unauthorized people cannot use.
- System must secure from unauthorized people.
- Manager must see the previous buyer record by using authentications.

3.3 Use Cases:

This section contains use cases of the store management system.

Use case user login:

Description:

Admin must be login to access this system.

Actor:

Admin

Precondition

Admin must be registered.

Basic flow:

- User enters his/her username and password.
- System check this data with database
- If username and password is correct.

Then user access our system

Else

User cannot access our system.

Alternative flow:

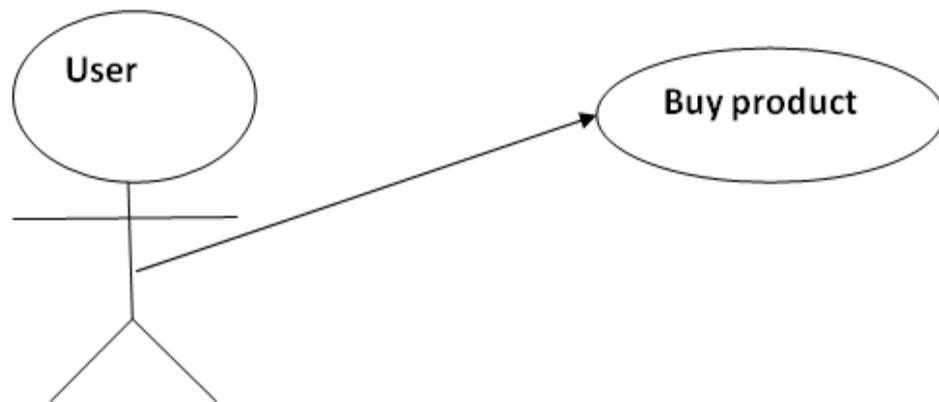
There is no alternative flow.

Post condition:

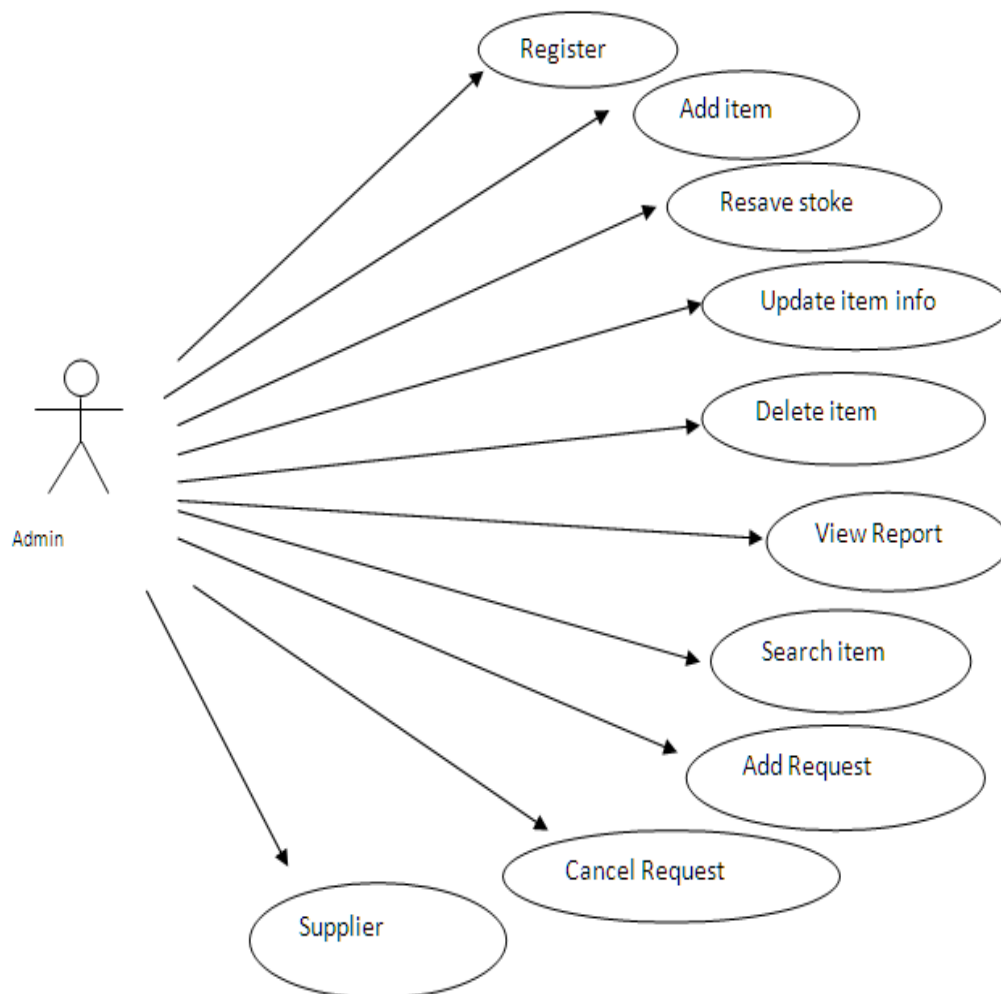
User should be login.

Exception:

Username and password is wrong.



3.3.2 User case:



Manager login	With username and password.
Modify and customer record	Manager can modify customer record and also can delete it.
Modify inventory record	Manager can update stock.
Check daily sale	As customer info is stored with date we can also check sale.
Buy product	Customer can buy product.
Generate bill	Manager will put all customer data in form and system will generate bill.
Customer detail	Every customer detail will store in database table.

3.4 Classes / Objects:

This section contains major classes of the store management system.

3.4.1<Admin>

3.4.1.1 Username, Password.

3.4.1.2 Admin login is for interaction with system

3.4.2<Buyer >

3.4.2.1 Name, CNIC, Address, City, Cell Number.

3.4.2.2 for storing information of buyers

3.4.3<All grocery company tables>

3.4.3.1 Grocery stock

3.4.3.2 for storing grocery & stock

3.5 Non-Functional Requirements:

- Interface well design.
- Application is easy to understand.
- Application is easy to handle.
- Performance is very fast.
- Doesn't store incorrect data.
- Data loss can't acceptable.
- System must secure from unauthorized person.
- System must secure from exposures.

3.6 Logical Database Requirements:

- Database will valid for this project.
- Our database keeps all record of our system.
- User can't interface in database.
- There will be store 3tables in database in admin, authentication, customer record's product's record.

3.7 Design Constraints:

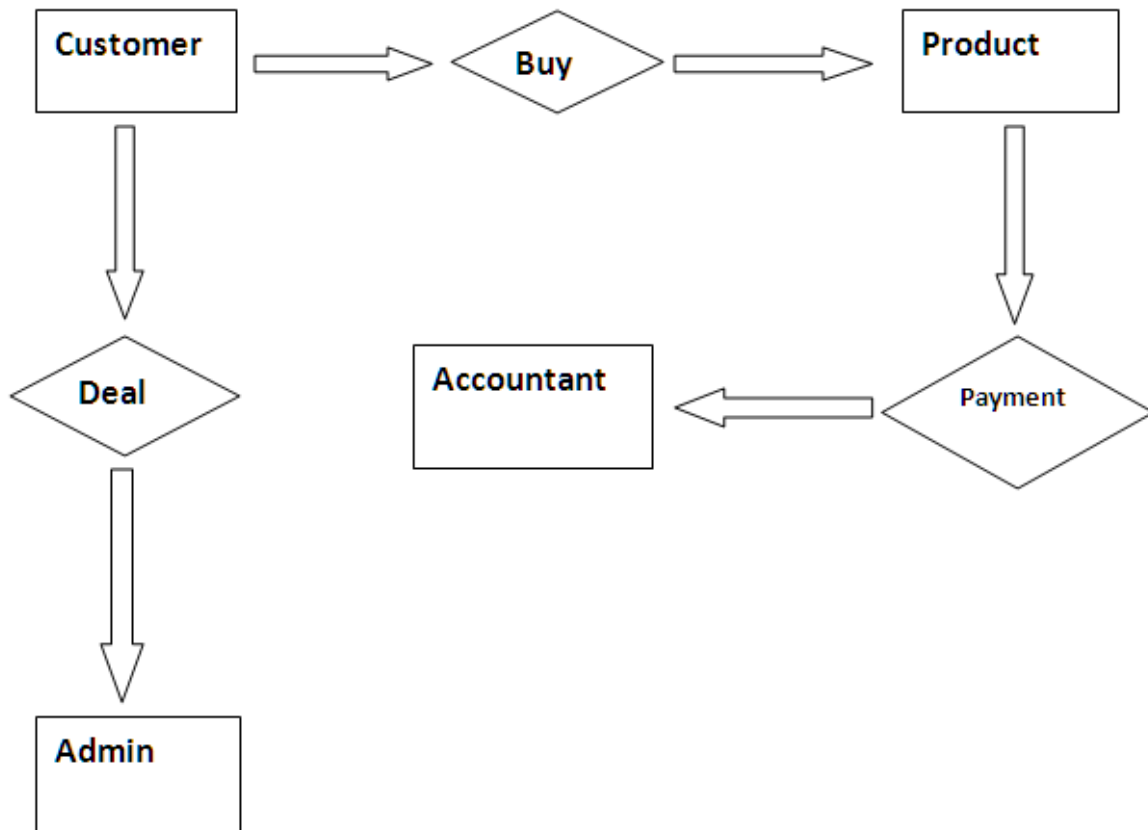
Specify design constraints that can be imposed by other standards, hardware limitations, etc.

3.7.1 Standards Compliance:

Specify the requirements derived from existing standards or regulations. They might include:

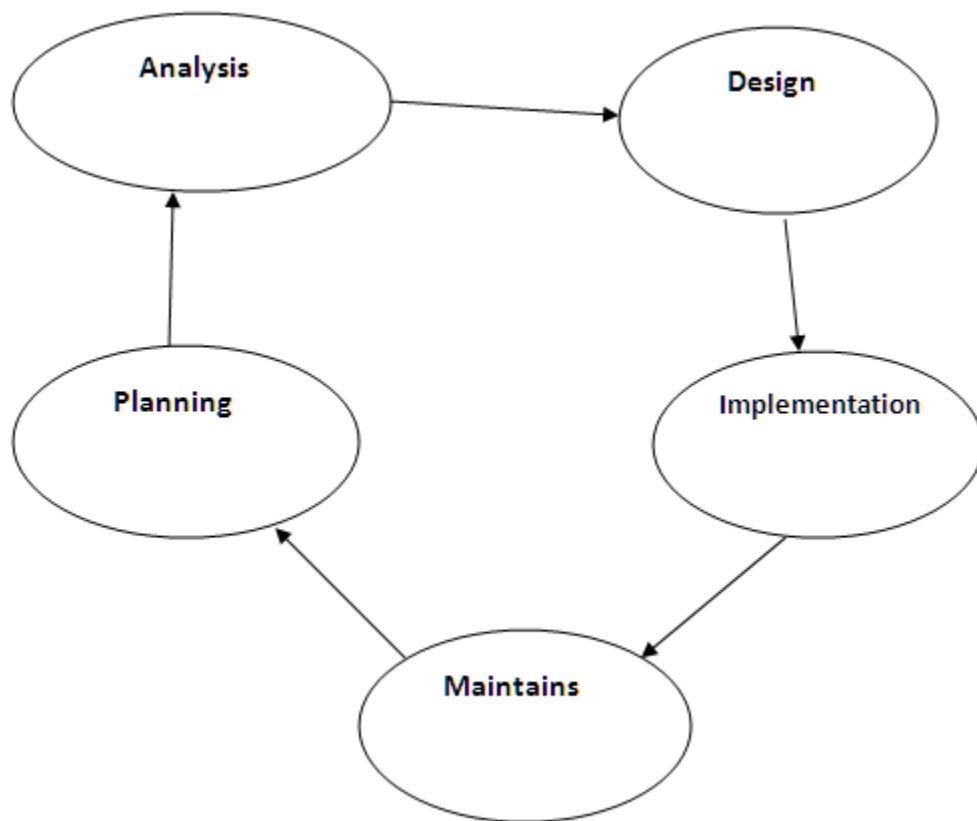
- Data naming
- Accounting procedures
- Audit Tracing

For example, this could specify the requirement for software to trace processing activity. Such traces are needed for some applications to meet minimum regulatory or financial standards. An audit trace requirement may, for example, state that all changes to a payroll database must be recorded in a trace file with before and after values.



CHAPTER 4: ANALYSIS MODELS

List all analysis models used in developing specific requirements previously given in this SRS. Each model should include an introduction and a narrative description. Furthermore, each model should be traceable the SRS's requirements.



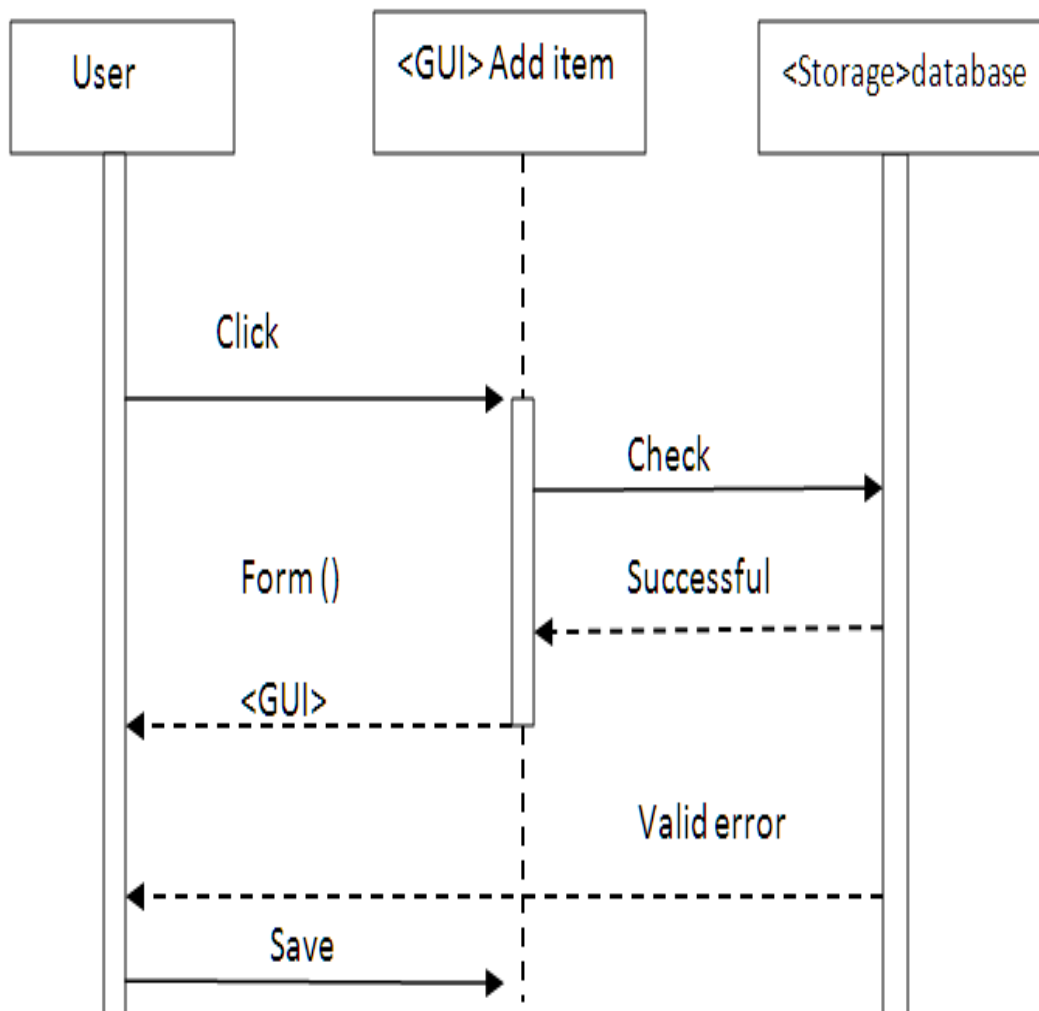
4.1 Sequence Diagrams:

Sequence diagrams are interaction diagrams that detail how operations are carried out

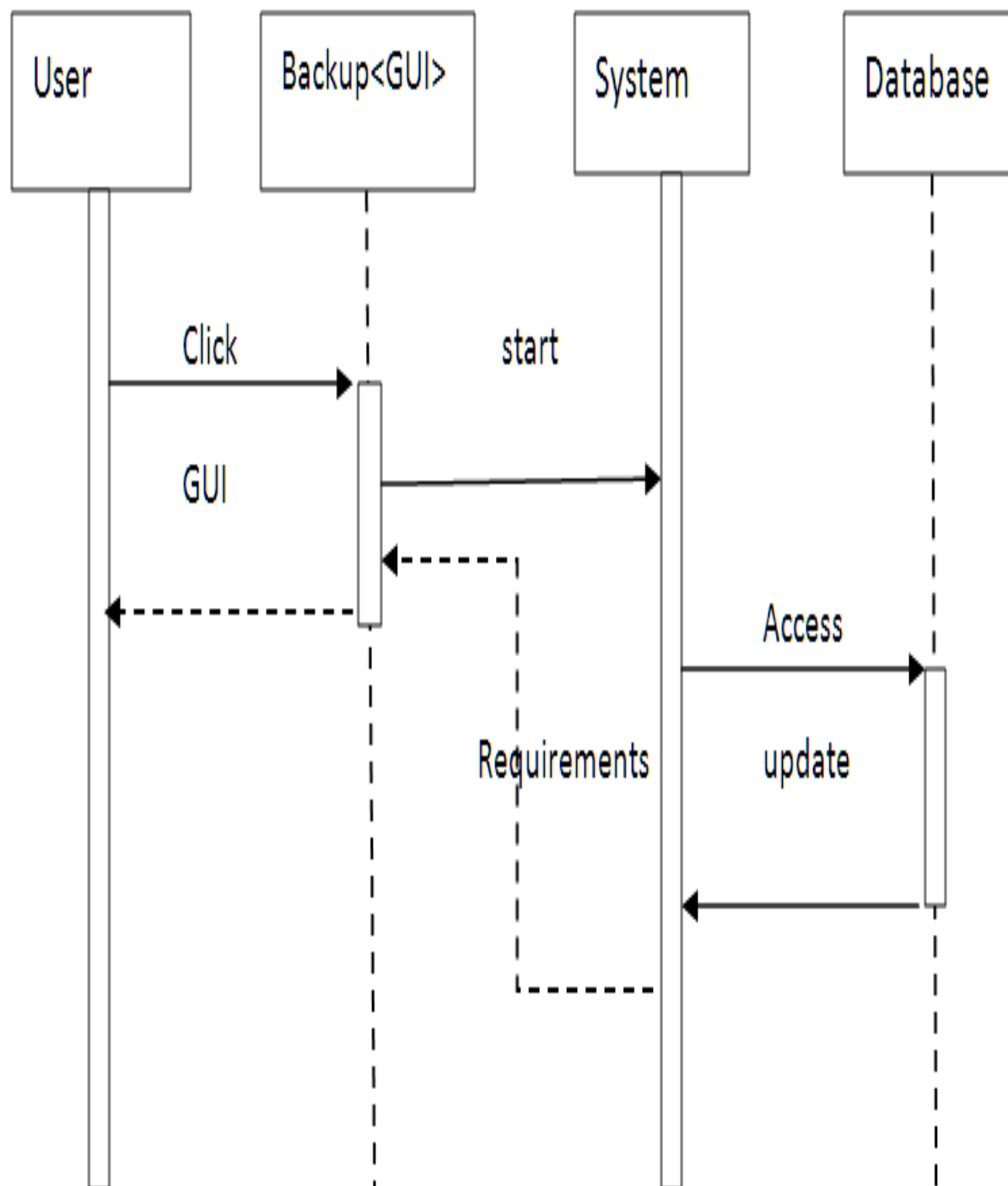
- Interaction diagrams model important runtime interaction between the parts that make up the system.
- A sequence diagram is made up of a collection of participants.

- Class or object- each class in the interaction is represented by its name icon along the top of the diagram.

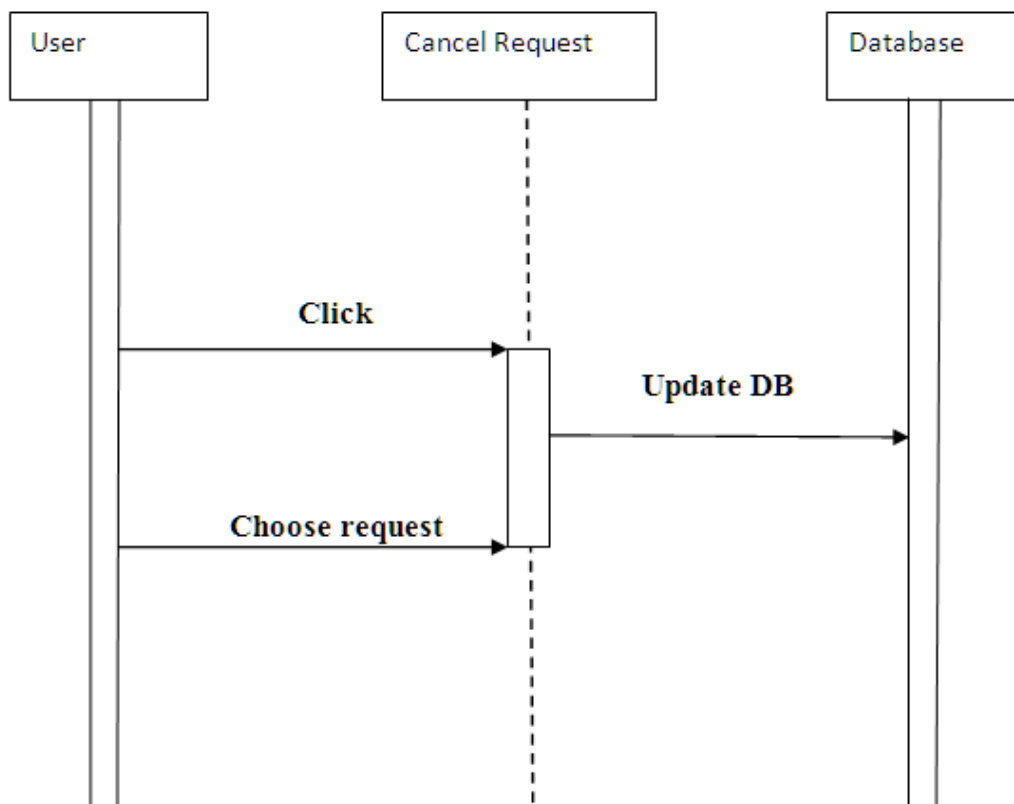
Add item:



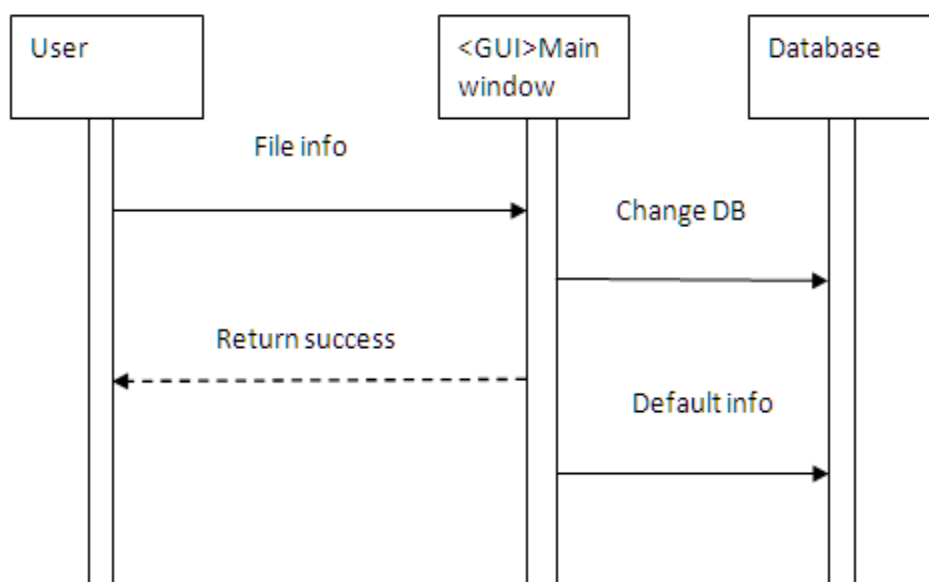
Sequence diagram for backup:



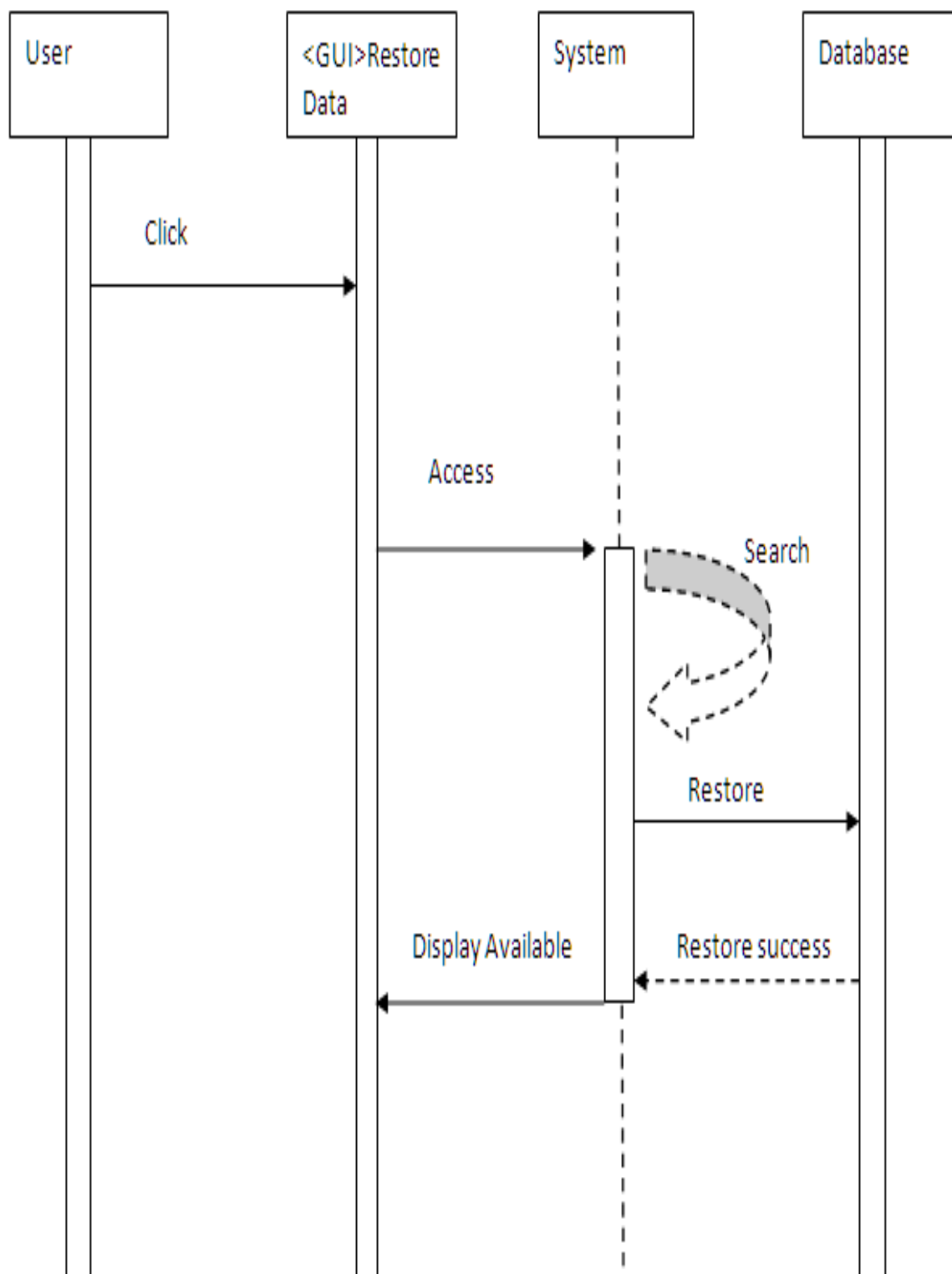
Sequence diagram for cancel request:



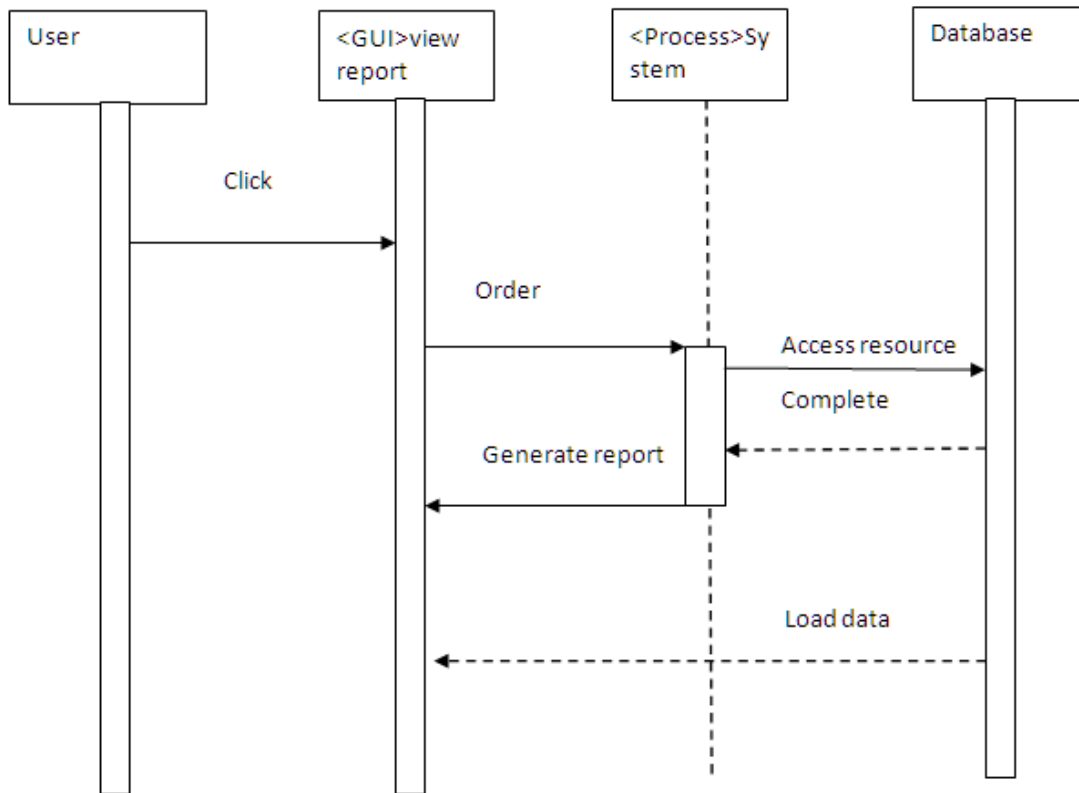
Sequence diagram for manage:



Sequence diagram for restore data:



Sequence diagram for view Report:



4.2 Activity diagram:

Activity diagram is graphical representation of workflows of stepwise activity and action with support for choice, iteration and concurrency. Activity diagram are often used in business process modeling. They can also describe the steps in a use case diagram. Activities modeled can be sequential and concurrent. In both cases an activity diagram will have a beginning (initial state) and an end (a final state).

Activity diagram are used for

- 1) Documenting existing process
- 2) Analysis new process concept
- 3) Finding reengineering opportunities.

Basic Activity Diagram Notations and Symbols:

Initial state and ending point:

A small filled circle followed by an arrow represents the initial action state or the start point for any activity diagram.



Activity:

An action state represents the non-interrupted action of objects. You can draw an action state in smart draw using a rectangle with round corners.



Action flow:

Action flows, also called edge and path, illustrate and transition from one action state to another. They are usually drawn with an arrowed line.



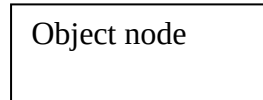
Activity final Node:

Stop all control flows and object flows in an activity.



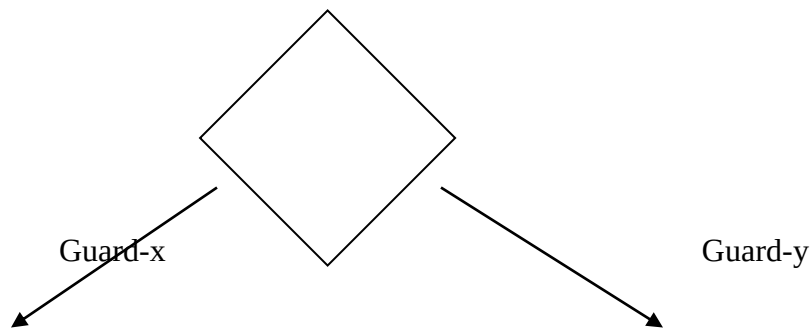
Object node:

Represent an object that is connected to a set of object flow.



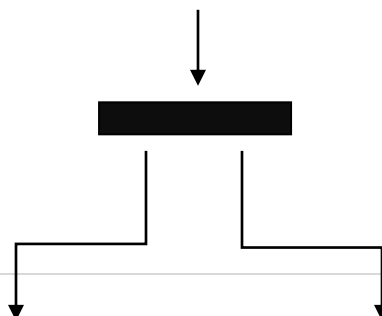
Decision Node:

Represent a test condition to ensure that the control flow or object flow only goes down one path.



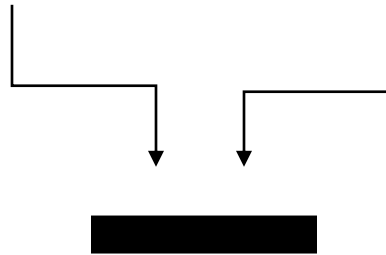
Fork Node:

Fork node is a control node that has one incoming edge and multiple outgoing edges and is used to split incoming flow into multiple concurrent flows.

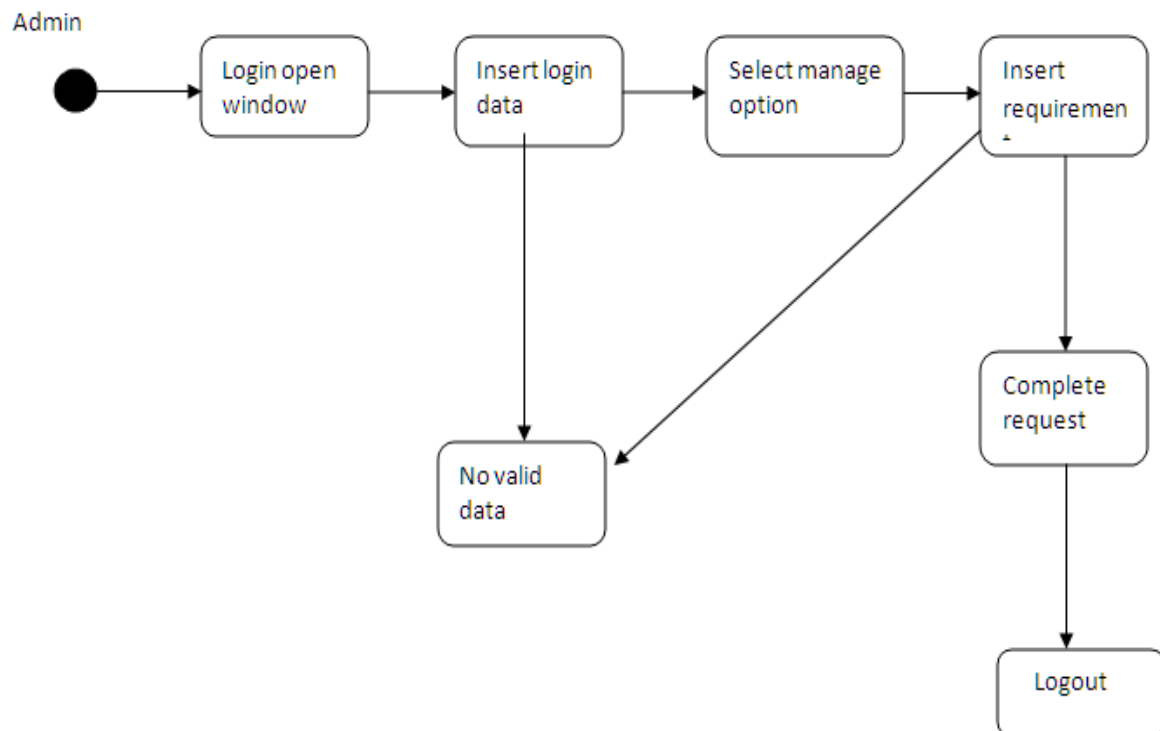


Join Node:

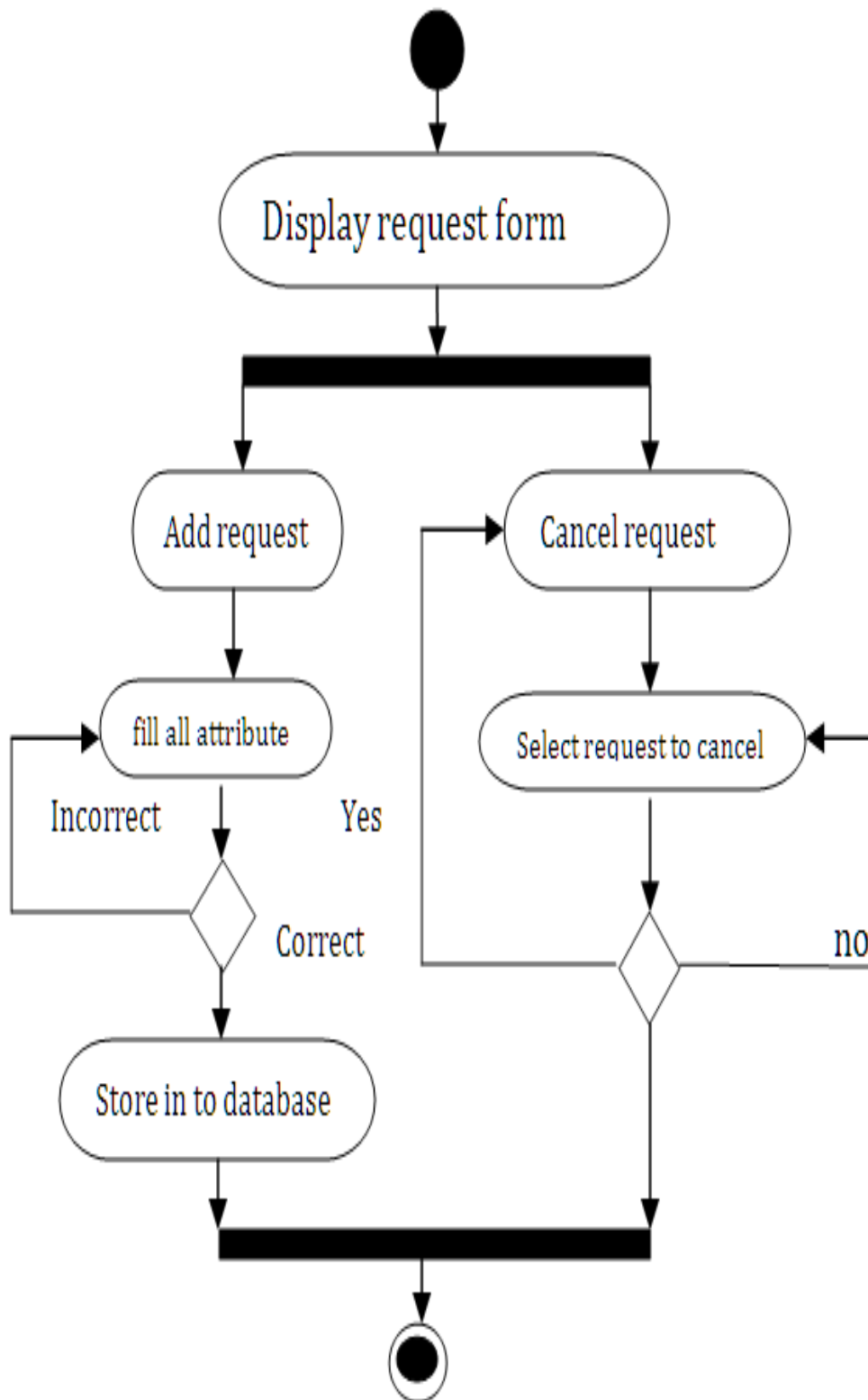
Bring back together a set of parallel or concurrent flows of activities.



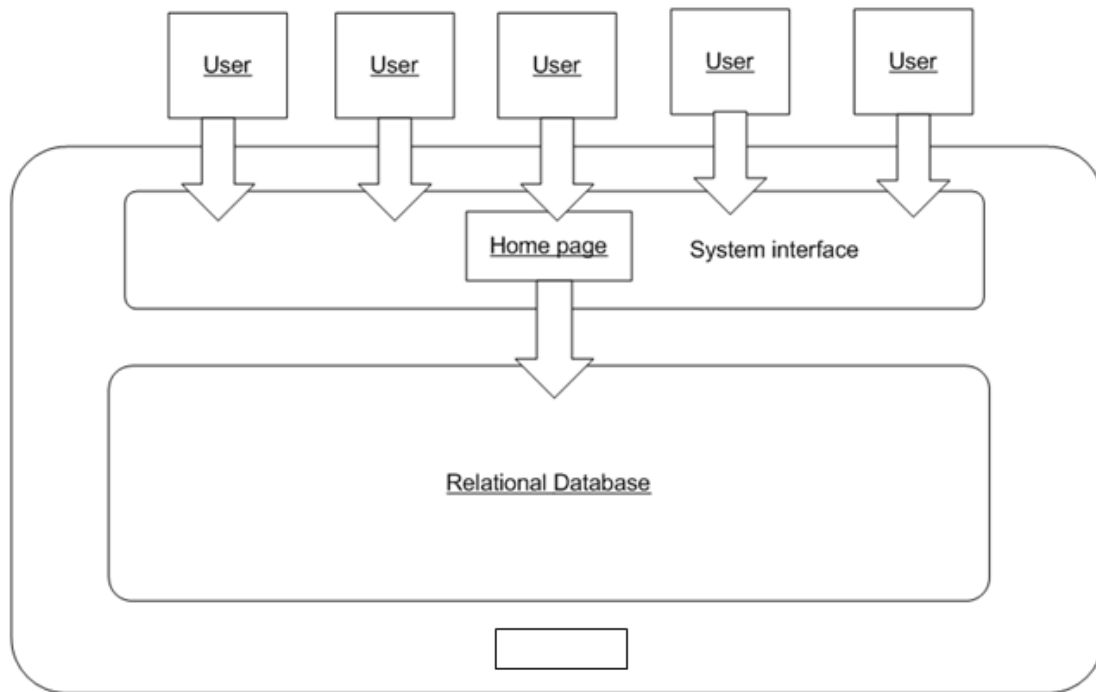
Activity Diagram:



Activity diagram for add/cancel request:



Block Diagram:



4.3 Data Flow Diagrams (DFD):

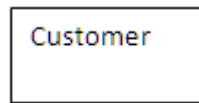
In a DFD, the arrows are actually data transfer between the elements, which are themselves parts of a system. DFD is visible part of diagrams of data flow model technique. The DFD is a diagram that consists principally of four symbols namely the external entity, the data flow, process, and the data store. Additionally, a physical flow can be shown on the DFD of the current system. A data flow diagram has no control flow; there are no decision rules and no loops.

Data flow diagram symbols:

Some symbols of DFD are as follows:

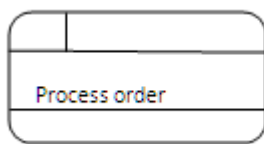
External Entity:

An external entity can represent a human, system and subsystem.



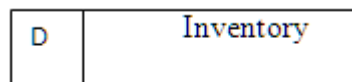
Process:

Set of operations that change values, form, or location of data.



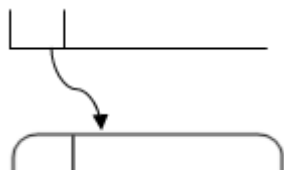
Data Store:

A data store represents the storage persistent data required and /or produced by the process.



Dataflow:

A data flow represents the flow of the information, with its direction represented by an arrow head that show at the end (s) of flow connector.

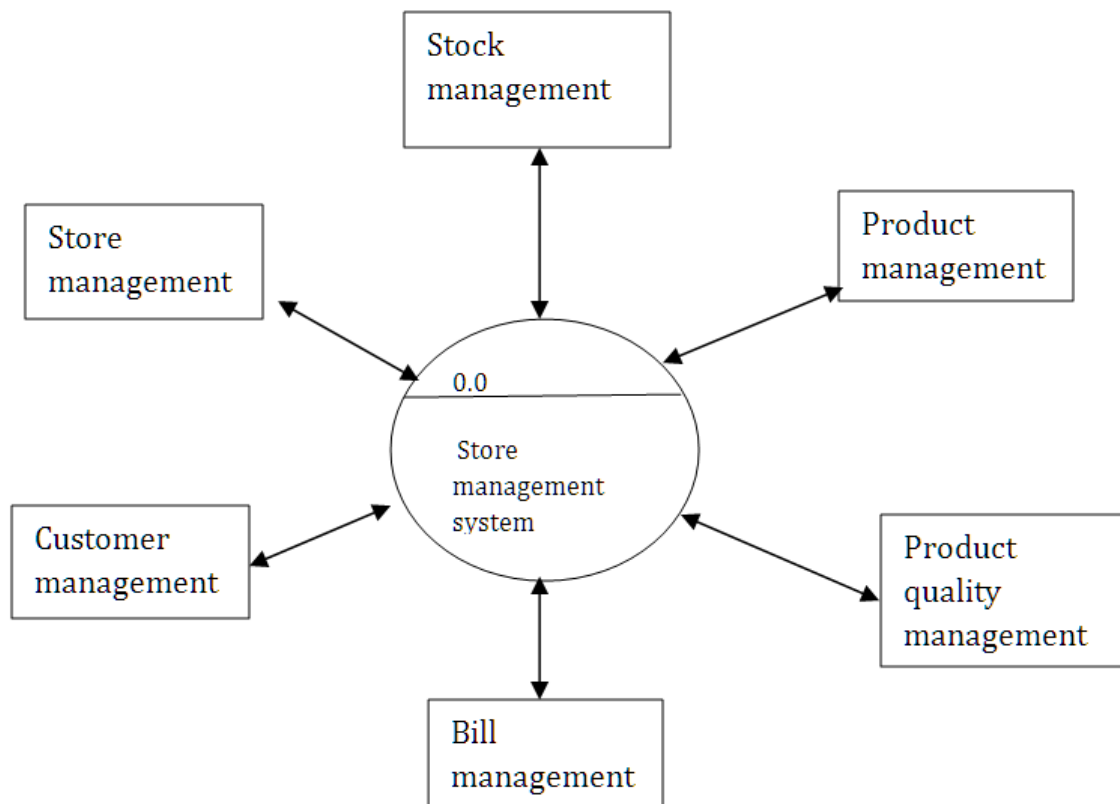


Following as:

- 1) 0-Level Diagram
- 2) Context Level Diagram
- 3) 1 – Level Diagram

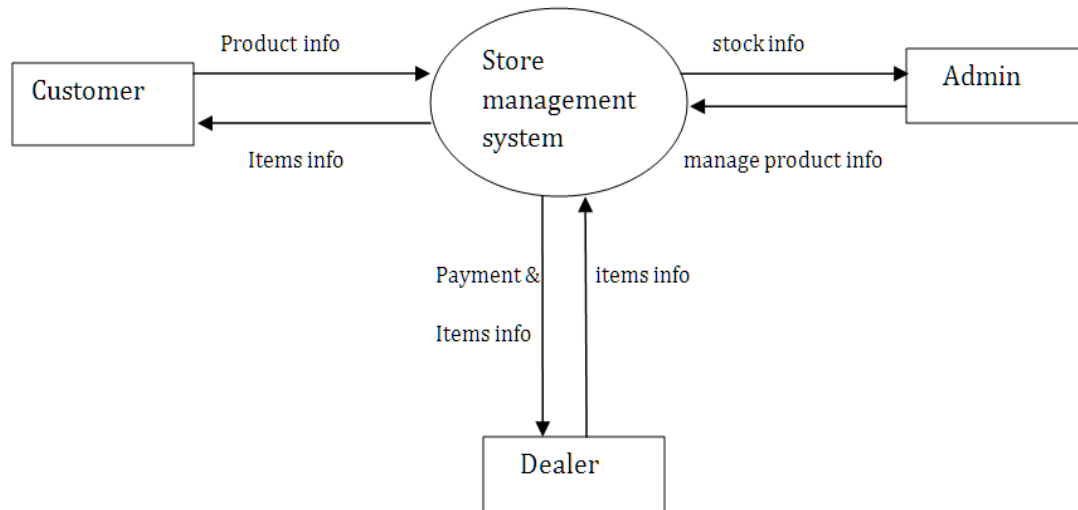
0-Level Diagram:

(o-level) data flow diagram may only contain one process node (process -0) that generalizes the function of the entry system in relationship to external entities. A(o-level)diagram is drawn in order to define and clarify the boundaries of the software system. It identifies the flows of information between the system and external entities.

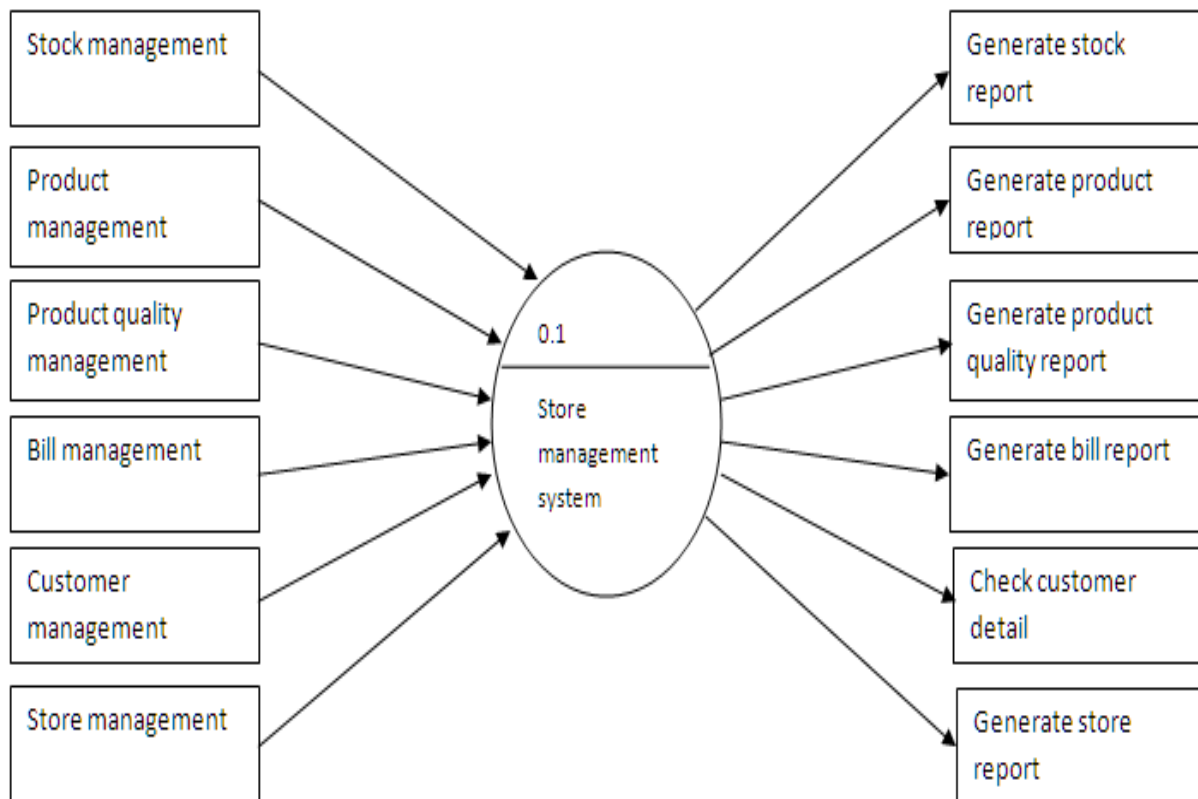


4.3.2 Context level diagram:

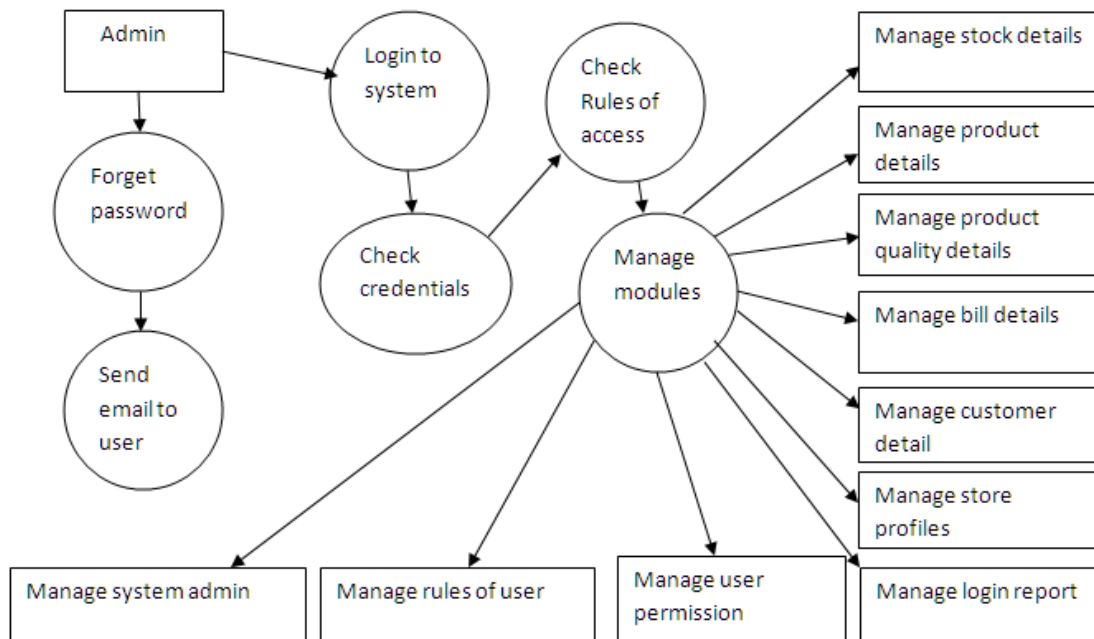
Context level is the top level dataflow diagram. It only contains one process node that generalizes the function of the entry system in relationship to external entities. A context level diagram is drawn in order to define and clarify the boundaries of the software system .It identifies the flows of information between the system and external entities.



1-level diagram:



4.3.4Detail level diagram:



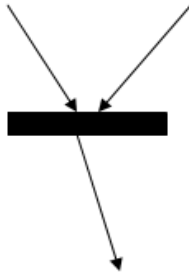
4.4 State-Transition Diagrams (STD)

In state transition diagram (STD) we represented that what actions leads are to change the state of the object

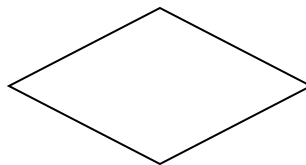
Transition:



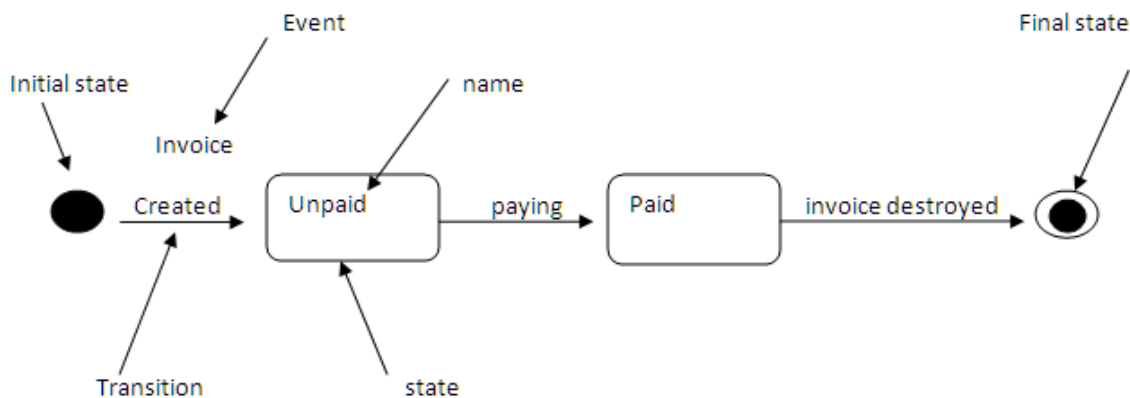
Transition Branch:

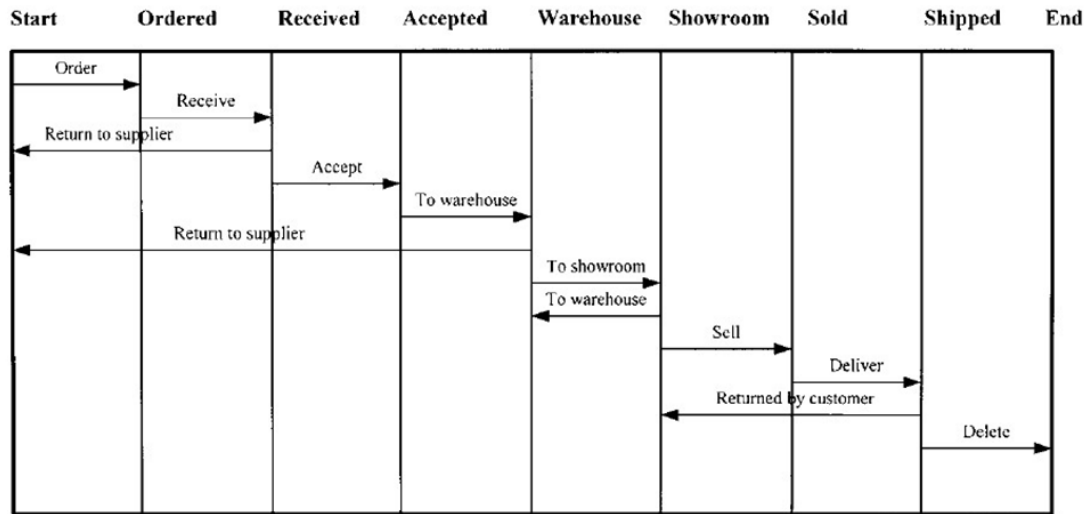


Decision symbol:



- A state is an abstraction of the attribute values and links of an object. Set of values are grouped together into a state according to properties that affect the gross behavior of the object.
- E.G... A bank is solvent or insolvent depending on whether its assets exceed its liabilities.
- A state corresponding to the internal between 2 events received by an object.
 - -A state separates 2 events.
 - -An event separates 2 states.





CHATER 5. SUPPORTING INFORMATION

Appendix A – Background Research on:

- Topic 1
- Topic 2
- Topic 3
-
- Topic n

Appendix B – Data Dictionary